$$= \lim_{h\to 0}\frac{f(0-h)-f(0)}{0-h-0},\ x=0-h$$

$$= \lim_{h\to 0}\frac{f(-h)-f(0)}{-h}$$

$$= \lim_{h\to 0}\frac{|-h|-|0|}{-h}$$

$$= \lim_{h\to 0}\frac{|-h|}{-h}$$

$$= \lim_{h\to 0}\frac{h}{-h}$$

$$= \lim_{h\to 0}(-1) = -1$$

$$R[f'(0)] = \lim_{x\to 0^+}\frac{f(x)-f(0)}{x-0}$$

$$= \lim_{h\to 0}\frac{f(0+h)-f(0)}{0+h-0},\ x=0+h$$

$$= \lim_{h\to 0}\frac{f(h)-f(0)}{h}$$

$$= \lim_{h\to 0}\frac{|h|-|0|}{h}$$

$$= \lim_{h\to 0}\frac{|h|}{h}$$

$$= \lim_{h\to 0}\frac{h}{h}$$

$$= \lim_{h\to 0}1 = 1$$

Here, $L[f'(0)] \neq R[f'(0)]$

$\therefore$ $f(x)$ is not differentiable at $x=0$

Example 5.30

Show that $f(x) = x^2$ differentiable at $x=1$ and find $f'(1)$.

Solution

$$f(x) = x^2$$

$$L[f'(1)] = \lim_{x\to 1^-}\frac{f(x)-f(1)}{x-1}$$

$$= \lim_{h\to 0}\frac{f(1-h)-f(1)}{1-h-1},\ x=1-h$$

$$= \lim_{h\to 0}\frac{(1-h)^2-1}{-h}$$

$$= \lim_{h\to 0}\frac{1+h^2-2h-1}{-h}$$

$$= \lim_{h\to 0}(-h+2) = 2$$

$$R[f'(1)] = \lim_{x\to 1^+}\frac{f(x)-f(1)}{x-1}$$

$$= \lim_{h\to 0}\frac{f(1+h)-f(1)}{1+h-1},\ x=1+h$$

$$= \lim_{h\to 0}\frac{(1+h)^2-1}{h}$$

$$= \lim_{h\to 0}\frac{1+h^2+2h-1}{h}$$

$$= \lim_{h\to 0}(h+2) = 2$$

Here, $L[f'(1)] = R[f'(1)]$

$\therefore$ $f(x)$ is differentiable at $x = 1$ and $f'(1) = 2$

5.2.12 Differentiation from first principle

The process of finding the derivative of a function by using the definition that

$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ is called the differentiation from first principle. It is convenient to write $f'(x)$ as $\frac{dy}{dx}$.

5.2.13 Derivatives of some standard functions using first principle

1. For $x \in R, \frac{d}{dx}(x^n) = n\,x^{n-1}$

Proof:

Let $f(x) = x^n$

$\therefore$ $f(x+h) = (x+h)^n$

$$\frac{d}{dx}(f(x)) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

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